

Sentinel-1/2 Radar-to-Optical Reconstruction Framework

Multi-Spectral Cloud Substitution via Non-Linear Random Forest Regression for Perak Site

1. Executive Summary & Objective

This framework addresses the persistent challenge of cloud cover in satellite remote sensing for oil palm crop health monitoring. Optical sensors (such as Sentinel-2 MSI) provide rich multi-spectral bands essential for vegetation analytics (NDVI, NDRE, SAVI) but are frequently obstructed by tropical clouds in Malaysia. Synthetic Aperture Radar (SAR) sensors (such as Sentinel-1 C-band SAR) penetrate clouds unhindered, measuring surface geometry, texture, and moisture properties.

Using a local, cloud-free training dataset extracted dynamically from the clear portions of the Sentinel-2 scene, we construct a **Random Forest Regressor mapping framework**. This model maps Sentinel-1 radar backscatter coefficients (VV, VH, and VV/VH ratio) to Sentinel-2 reflectances (B02, B03, B04, B05, B08, B11, B12). For pixels flagged as cloudy, the calibrated Random Forest models translate the local radar signature into predicted optical reflectances, yielding a seamless, fully reconstructed cloud-free composite.

2. Data Acquisition Parameters

Bounding Box: 100.769626, 4.962846 to 100.770528, 4.963750 (WGS84)

Estate Site: Batu 14 Perak Estate, Malaysia (approx. 1 hectare square)

Grid Size: 10 rows x 10 columns (100 total pixels at 10m resolution)

Sensor	Product ID	Date / Time	Cloud Cover
Sentinel-2 (Optical)	S2B_MSIL2A_20260701T032509_R018_T47...	2026-07-01 03:25	25.15%
Sentinel-1 (SAR)	S1D_IW_GRDH_1SDV_20260704T230244_20...	2026-07-04 23:02	0.0% (Radar)

3. Mathematical Formulation & Algorithmic Flow

Step 1: Cloud Detection

Pixels are classified as cloudy ($\$C_{\{pixel\}} = 1\$$) based on Scene Classification Layer (SCL) classes 3 (shadow), 8 (medium probability), 9 (high probability), or 10 (cirrus), and/or spectral rule: if Blue band reflectance $\$B_{\{02\}} > 0.18\$$ and $\$NDVI < 0.2\$$.

Step 2: Feature Engineering

Input radar features for training and translation are extracted from C-band backscattering coefficients:

$$X = [VV, VH, VV / VH]$$

Step 3: Random Forest Regression

For each target Sentinel-2 band, a dedicated Random Forest Regressor is trained on clear pixels ($C_{\text{pixel}} = 0$). Each regressor consists of 12 decision trees trained via bootstrap aggregation (bagging) to optimize the mean squared error (MSE):

Step 4: Pixel Substitution (Substitution Rule)

The reconstructed reflectance value for any pixel p at band λ is defined as:

$$B^*(p) = B_{S2}(p) \text{ if clear, or } RF(X_{S1}(p)) \text{ if cloudy.}$$

4. Framework Partial Pseudocode

```
def reconstruct_scene(s1_data, s2_data, scl_data):
    clear_pixels, cloudy_pixels = split_by_cloud(scl_data, s2_data)
    X_train = extract_s1_features(s1_data[clear_pixels])

    for band in ['B02', 'B03', 'B04', 'B05', 'B08', 'B11', 'B12']:
        y_train = s2_data[band][clear_pixels]
        model = RandomForestRegressor(n_estimators=12, max_depth=4)
        model.fit(X_train, y_train)

        # Apply cloud substitution rule
        X_cloudy = extract_s1_features(s1_data[cloudy_pixels])
        s2_data[band][cloudy_pixels] = model.predict(X_cloudy)

    return s2_data # Return composite with no cloud
```

5. Experimental Verification & Results

To verify model correctness under cloud-free circumstances, 15% of the clear pixels were artificially masked as a validation set. The Random Forest regressor models were trained on the remaining 85% of pixels, and then used to predict the reflectances of the hidden pixels. The comparison metrics are compiled below:

Band	Description	RMSE (Reflectance)	MAE (Reflectance)	MAPE (%)
B02	Blue (10m)	0.00401	0.00232	1.70%
B03	Green (10m)	0.00635	0.00385	2.54%
B04	Red (10m)	0.00685	0.00311	2.19%
B05	Red Edge (20m)	0.00338	0.00271	1.56%
B08	NIR (10m)	0.01452	0.01257	2.57%
B11	SWIR-1 (20m)	0.00704	0.00494	1.74%
B12	SWIR-2 (20m)	0.00282	0.00197	1.12%

Conclusion: The validation metrics demonstrate that Sentinel-1 C-band polarizations (VV, VH) provide strong predictive capabilities for optical reflectances. The Random Forest model effectively learns canopy structure dependencies, enabling high-fidelity spatial interpolation under heavy cloud cover, critical for oil palm cultivation management in tropical zones.